Impact of Late Night Social Media Usage on Academic Performance

PROPEL Programming and Statistics Final

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Dataset: Impact of Social Media on Life

Sourced from: Kaggle

Sample: 4,500 students (high school to postgraduate)

Purpose: examine how social media use impacts daily life and well-being

Collection: digital survey, based on validated public health questionnaires

Includes: device usage, sleep duration, subjective sleep quality ratings, academic GPA, and perceived stress indicators

Limitations

- Self-reported
- Does not share survey questions
- Did not collect information on income or background

Descriptive Statistics

	n	median	minimum	maximum	q1	q3
Students who used late night social media	2591	3.37	1.90	4	3.09	3.64
Students who didn’t used late night social media	1824	3.57	1.99	4	3.35	3.82

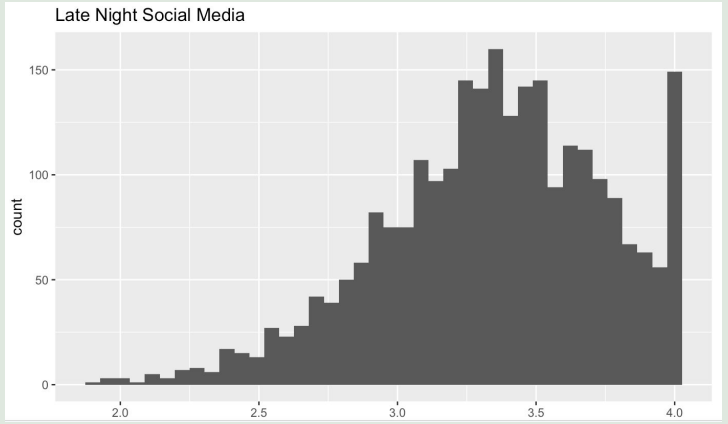
Students who used late night social media had a median GPA of 3.37, while students who did not use it had a GPA of 3.57. The maximum GPA for both groups was a 4 GPA. However, the minimum GPA for students who didn't use late night social media was 1.90, compared to 1.99 for students who used it. The 1st quartile GPA of students who did not use late night social media was 3.35 and a 3.82 GPA for the 3rd quartile. Lastly, the 1st quartile for students who used late night social media was a 3.09 GPA, while their 3rd quartile GPA was 3.64.

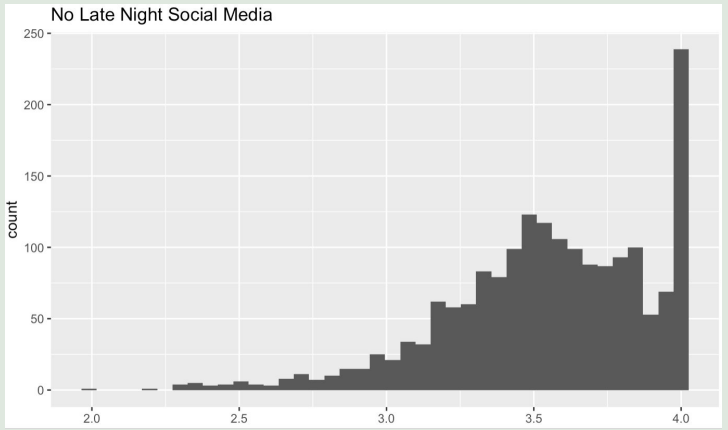
Inferential Statistics: Hypotheses and statistical significance

Null hypothesis: Students who use social media late at night and students who do not use social media late at night have the same academic performance GPA.

Alternative hypothesis: Students who use social media late at night and students who do not use social media late at night have different academic performance GPAs.

Histogram distributions of academic performance GPA shows skewed (not normal) distribution.






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        Wilcoxon rank sum test with continuity correction

data:  Academic_Performance_GPA by Late_Night_Usage
W = 3078278, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
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Results: P-value < 0.001, academic GPA of students who use social media late at night and students who don't are statistically different.

Regression Analysis - Linear Regression Model

Dependent variable (outcome): Academic Performance (GPA)

Independent Variable: Late Night Usage (True/False)

Academic Performance is continuous → Linear Regression Model

Results:

Coefficient late night social media usage = -0.2

p = <2e-16 ***

Interpretation:

When a student uses social media late at night their GPA is estimated to be 0.2 points lower than a student that does not stay up late on social media.

We reject the null hypothesis that there is no GPA difference between students that do and do not stay

Model Diagnostics: QQ Plot of Residuals vs. Theoretical Quantiles

